Naive Bayes Classification and Variable Elimination

Two uses of probability in AI: assigning a class to an observation and computing a probability in a network of related variables.

Probability tools used in both algorithms

Quantity	Meaning
Prior, $\mathsf{P}(H)$	Probability assigned to a hypothesis before observing the evidence.
Likelihood, $\mathsf{P}(E \mid H)$	Probability of the evidence when the hypothesis is true.
Posterior, $\mathsf{P}(H \mid E)$	Probability of the hypothesis after observing the evidence.
Marginal	Probability of a variable after summing over other variables.

For mutually exclusive and exhaustive classes c, Bayes' theorem gives

$$\mathsf{P}(C = c \mid E) = \frac{\mathsf{P}(E \mid C = c)\mathsf{P}(C = c)}{\sum_{c'} \mathsf{P}(E \mid C = c')\mathsf{P}(C = c')}.$$

For a binary variable X, the two probabilities add to one:

$$\mathsf{P}(X = 0) = 1 - \mathsf{P}(X = 1).$$

The law of total probability sums over all values of an unobserved variable:

$$\mathsf{P}(B = b) = \sum_{a} \mathsf{P}(B = b \mid A = a)\mathsf{P}(A = a).$$

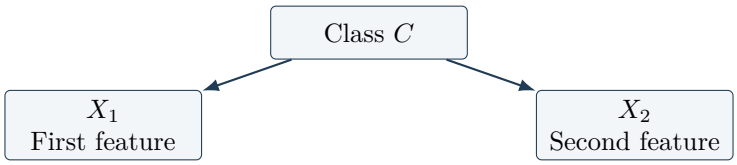
1. Naive Bayes: model and algorithm

CHAPTER SECTION 5.4 | APPLICATION: SPAM DETECTION

Let C be the class and $X_1, \ldots, X_d$ be the features. Naive Bayes assumes that the features are **conditionally independent given the class**:

$$\mathsf{P}(X_1 = x_1, \ldots, X_d = x_d \mid C = c) = \prod_{i=1}^{d} \mathsf{P}(X_i = x_i \mid C = c).$$

This does not require the features to be independent without conditioning on C.



Training: estimate the probabilities from labelled data

Let N be the total number of examples, N_c the number in class c, and $N_{i,v,c}$ the number in class c with feature value $X_i = v$. For categorical features, estimate

$$\mathsf{P}(C = c) = \frac{N_c}{N}, \qquad \mathsf{P}(X_i = v \mid C = c) = \frac{N_{i,v,c}}{N_c}.$$

For example, consider eight labelled emails with these counts:

Count	Spam	Legitimate
Number of emails	4	4
Emails containing “free”	3	1
Emails containing “win”	3	1

The class priors are both $4/8 = 0.5$. The probabilities of each word being present are $3/4 = 0.75$ in spam and $1/4 = 0.25$ in legitimate emails.

Prediction: compute and compare the scores

For observed features $\boldsymbol{x} = (x_1, \ldots, x_d)$, define

$$S(c) = \mathsf{P}(C = c) \prod_{i=1}^{d} \mathsf{P}(X_i = x_i \mid C = c), \qquad \widehat{c} = \arg\max_c S(c).$$

The scores are unnormalized. Under the model, $\mathsf{P}(C = c \mid \boldsymbol{x}) = S(c) / \sum_{c'} S(c')$.

Algorithm: Naive Bayes classification

1. **Train:** estimate the class priors and feature probabilities.
2. **Score:** for each class, start with its prior and multiply by the likelihood of every observed feature value.
3. **Classify:** return the class with the largest score.
4. **If needed:** normalize the scores to obtain posterior probabilities.

2. Worked problem: classify an email

An email contains both "free" and "win". Should it be classified as spam? Use the following supplied probabilities; this is a different model from the small training illustration on page 2.

Define $C = 1$ for spam and $C = 0$ for legitimate email. Let $X_1 = 1$ mean "free" is present and $X_2 = 1$ mean "win" is present. A value of 0 means that the corresponding word is absent.

Probability	**Spam:** $C = 1$	**Legitimate:** $C = 0$
$\mathsf{P}(C)$	0.4	0.6
$\mathsf{P}(X_1 = 1 \mid C)$	0.7	0.2
$\mathsf{P}(X_2 = 1 \mid C)$	0.8	0.3

Step 1: use conditional independence

$$\mathsf{P}(X_1 = 1, X_2 = 1 \mid C = 1) = (0.7)(0.8) = 0.56,$$
$$\mathsf{P}(X_1 = 1, X_2 = 1 \mid C = 0) = (0.2)(0.3) = 0.06.$$

Step 2: include the class priors

$$S(1) = (0.4)(0.56) = 0.224,$$
$$S(0) = (0.6)(0.06) = 0.036.$$

Since $0.224 > 0.036$, the predicted class is $\widehat{c} = 1$.

Step 3: normalize if a probability is requested

The normalizing constant is $0.224 + 0.036 = 0.260$. Therefore,

$$\mathsf{P}(C = 1 \mid X_1 = 1, X_2 = 1) = \frac{0.224}{0.260} \approx 0.8615,$$

$$\mathsf{P}(C = 0 \mid X_1 = 1, X_2 = 1) = \frac{0.036}{0.260} \approx 0.1385.$$

Answer and interpretation

Classify the email as **spam**. The posterior probability of spam is approximately **86.15% under the Naive Bayes model**. The two posterior probabilities add to one.

What if a word is absent?

In this binary-feature model, absence is also an observed feature value. For example, if "free" is present but "win" is absent, use

$$\mathsf{P}(X_2 = 0 \mid C = 1) = 1 - 0.8 = 0.2, \quad \mathsf{P}(X_2 = 0 \mid C = 0) = 1 - 0.3 = 0.7.$$

Thus $S(1) = 0.4(0.7)(0.2) = 0.056$ and $S(0) = 0.6(0.2)(0.7) = 0.084$. This email is classified as **legitimate**.

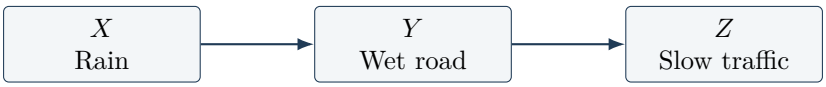
Common mistake

The spam score 0.224 is not the posterior probability of spam. Normalize the two scores to obtain that probability. Always include the prior when computing each score.

3. Variable Elimination: model and algorithm

A **Bayesian network** represents a joint probability distribution using a directed acyclic graph and conditional probability tables. Each node is a variable. Arrows specify which variables appear as parents in its conditional probability table; an arrow alone does not establish a causal relationship.

Consider a simple model for rain, road conditions, and traffic:



The model assumes that Z is conditionally independent of X given Y. Consequently, the joint distribution factorizes as

$$\mathsf{P}(X, Y, Z) = \mathsf{P}(X)\mathsf{P}(Y \mid X)\mathsf{P}(Z \mid Y).$$

The three probability tables are called **factors**. A factor associates a number with each assignment of the variables in its scope.

What does it mean to eliminate a variable?

Suppose the query is $\mathsf{P}(Z)$. The values of X and Y are not observed, so both must be accounted for:

$$\mathsf{P}(Z) = \sum_y \sum_x \mathsf{P}(X = x)\mathsf{P}(Y = y \mid X = x)\mathsf{P}(Z \mid Y = y).$$

We can first combine the factors involving X and sum out X:

$$f(Y) = \sum_x \mathsf{P}(X = x)\mathsf{P}(Y \mid X = x).$$

Then combine $f(Y)$ with the remaining factor and sum out Y:

$$\mathsf{P}(Z) = \sum_y f(Y = y)\mathsf{P}(Z \mid Y = y).$$

The new factor $f(Y)$ retains the effect of X. We have not ignored X.

Algorithm: Variable Elimination

1. **Form factors:** write the probability tables from the network.
2. **Enter evidence:** if a variable is observed, restrict every factor containing it to its observed value.
3. **Choose an order:** list the hidden variables that are neither query variables nor evidence variables.
4. **Eliminate:** for each hidden variable, multiply all current factors containing it, sum over its possible values, and replace those factors with the resulting factor.
5. **Finish:** multiply the remaining factors. Normalize when computing a conditional distribution.

Meaning of elimination

Multiply the relevant factors, then sum over the variable's values. The algorithm gives an exact result for the supplied model. An elimination order can change the sizes of intermediate factors and the amount of work, but it does not change the mathematical answer.

4. Worked problem: predict slow traffic

Use the network $X \to Y \to Z$ from page 4. All variables are binary: 1 means that the event occurs and 0 means that it does not. The prior is $\mathsf{P}(X = 1) = 0.6$, so $\mathsf{P}(X = 0) = 0.4$.

X	$\mathsf{P}(Y = 1 \mid X)$	$\mathsf{P}(Y = 0 \mid X)$
1	0.8	0.2
0	0.3	0.7

Y	$\mathsf{P}(Z = 1 \mid Y)$	$\mathsf{P}(Z = 0 \mid Y)$
1	0.9	0.1
0	0.2	0.8

Query: compute $\mathsf{P}(Z = 1)$, the probability of slow traffic. **Elimination order:** X, followed by Y.

Step 1: eliminate rain, X

Multiply $\mathsf{P}(X)$ and $\mathsf{P}(Y \mid X)$ and sum over X:

$$\begin{aligned} f(Y = 1) &= \mathsf{P}(X = 1)\mathsf{P}(Y = 1 \mid X = 1) \\ &\quad + \mathsf{P}(X = 0)\mathsf{P}(Y = 1 \mid X = 0) \\ &= (0.6)(0.8) + (0.4)(0.3) = 0.48 + 0.12 = 0.60, \\ f(Y = 0) &= (0.6)(0.2) + (0.4)(0.7) = 0.12 + 0.28 = 0.40. \end{aligned}$$

Here $f(Y)$ is the marginal distribution of road wetness: the probability of a wet road is 0.60 and the probability of a dry road is 0.40.

Step 2: eliminate road wetness, Y

Multiply $f(Y)$ and $\mathsf{P}(Z \mid Y)$ and sum over Y:

$$\begin{aligned} \mathsf{P}(Z = 1) &= f(Y = 1)\mathsf{P}(Z = 1 \mid Y = 1) \\ &\quad + f(Y = 0)\mathsf{P}(Z = 1 \mid Y = 0) \\ &= (0.60)(0.9) + (0.40)(0.2) \\ &= 0.54 + 0.08 = \boxed{0.62}. \end{aligned}$$

Similarly,

$$\mathsf{P}(Z = 0) = (0.60)(0.1) + (0.40)(0.8) = 0.06 + 0.32 = 0.38.$$

Answer and check

The probability of slow traffic is **62%**. The final distribution is $\mathsf{P}(Z = 1) = 0.62$ and $\mathsf{P}(Z = 0) = 0.38$; these add to one. No extra normalization is needed for this marginal query.

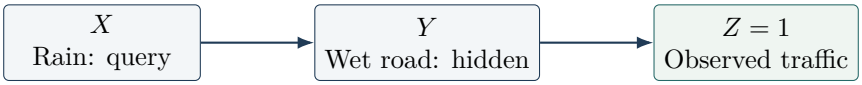
Track the factors

Stage	Current factors
Initially	$\mathsf{P}(X)$, $\mathsf{P}(Y \mid X)$, $\mathsf{P}(Z \mid Y)$
After eliminating X	$f(Y)$, $\mathsf{P}(Z \mid Y)$
After eliminating Y	$\mathsf{P}(Z)$

5. Extension: infer rain from observed traffic

POSTERIOR INFERENCE | OPTIONAL EXTENSION

Use the same network and probability tables. This time, slow traffic has been observed, so $Z = 1$ is **evidence**. Find $\mathsf{P}(X = 1 \mid Z = 1)$.



Only Y is eliminated. Keep X because it is the query variable.

Step 1: restrict the traffic factor to the evidence

Set $Z = 1$ in $\mathsf{P}(Z \mid Y)$. The resulting factor is

$$g(Y) = \mathsf{P}(Z = 1 \mid Y), \qquad g(1) = 0.9, \quad g(0) = 0.2.$$

Step 2: multiply and eliminate Y

Create a factor over X:

$$h(X) = \sum_y \mathsf{P}(Y = y \mid X) g(Y).$$

For each value of X,

$$h(1) = (0.8)(0.9) + (0.2)(0.2) = 0.72 + 0.04 = 0.76,$$
$$h(0) = (0.3)(0.9) + (0.7)(0.2) = 0.27 + 0.14 = 0.41.$$

These are likelihoods: $h(x) = \mathsf{P}(Z = 1 \mid X = x)$. They are not yet the posterior distribution of rain.

Step 3: multiply by the prior

The remaining factors give the unnormalized distribution $u(X) = \mathsf{P}(X)h(X)$:

$$u(1) = (0.6)(0.76) = 0.456,$$
$$u(0) = (0.4)(0.41) = 0.164.$$

Their sum is $0.456 + 0.164 = 0.620 = \mathsf{P}(Z = 1)$.

Step 4: normalize

$$\mathsf{P}(X = 1 \mid Z = 1) = \frac{0.456}{0.620} \approx \boxed{0.7355},$$
$$\mathsf{P}(X = 0 \mid Z = 1) = \frac{0.164}{0.620} \approx 0.2645.$$

Interpretation

The prior probability of rain was 60%. After observing slow traffic, the posterior probability is approximately **73.55% under this model**. Slow traffic is evidence supporting rain, but it does not prove that it rained.

6. Compare the methods and practise

	Naive Bayes	**Variable Elimination**
Main purpose	Predict a class from observed features.	Compute a marginal or posterior in a probabilistic model.
Model	Class priors and feature likelihoods; features conditionally independent given the class.	Factors determined by the network; dependencies are represented by its structure.
Main calculation	Multiply the prior and feature likelihoods for each class.	Multiply factors and sum over hidden variables.
Output	A predicted class; posterior probabilities if normalized.	An exact probability distribution for the supplied model.
Example	Spam or legitimate email.	Probability of traffic or rain.

How they relate

Naive Bayes is a particular probabilistic classification model. Variable Elimination is an inference algorithm. Both use probability factorization, and Variable Elimination can also be applied to a Naive Bayes network.

Short questions

1. State the conditional independence assumption of Naive Bayes.
2. Why can we compare class scores without first normalizing them?
3. What does it mean to sum out a binary variable?
4. In the query $\mathsf{P}(X \mid Z = 1)$ for $X \to Y \to Z$, identify the query, evidence, and hidden variables.

Problem A: Naive Bayes

Using the probability table on page 3, classify an email that contains "win" but does *not* contain "free". Calculate both scores and the posterior probability of spam.

Problem B: learn from counts

Using the eight-email training counts on page 2, classify a new email containing both words. Calculate the posterior probability of spam under that model.

Problem C: Variable Elimination

For the network $X \to Y \to Z$, suppose

$$\mathsf{P}(X = 1) = 0.5,$$
$$\mathsf{P}(Y = 1 \mid X = 1) = 0.8, \qquad \mathsf{P}(Y = 1 \mid X = 0) = 0.2,$$
$$\mathsf{P}(Z = 1 \mid Y = 1) = 0.9, \qquad \mathsf{P}(Z = 1 \mid Y = 0) = 0.1.$$

(a) Eliminate X, then Y, to compute $\mathsf{P}(Z = 1)$.

(b) **Extension:** if $Z = 1$ is observed, eliminate Y and compute $\mathsf{P}(X = 1 \mid Z = 1)$.

7. Answers and study reminders

Answers to the short questions

1. Given the class, the joint feature likelihood equals the product of the individual feature likelihoods.
2. All scores share the same positive normalizing denominator. Division by it does not change which score is largest.
3. Add the contributions for both values, 0 and 1. The resulting factor no longer contains that variable.
4. Query: X. Evidence: $Z = 1$. Hidden variable: Y.

Problem A

The observation is $X_1 = 0, X_2 = 1$. Use the complement for the absent word:

$$S(1) = (0.4)(1 - 0.7)(0.8) = 0.096,$$
$$S(0) = (0.6)(1 - 0.2)(0.3) = 0.144.$$

The prediction is **legitimate**. The posterior probability of spam is $0.096/(0.096 + 0.144) = \boxed{0.40}$.

Problem B

$$S(1) = (0.5)(0.75)(0.75) = 0.28125, \qquad S(0) = (0.5)(0.25)(0.25) = 0.03125.$$

The prediction is **spam**; its posterior probability is $0.28125/(0.28125 + 0.03125) = \boxed{0.90}$.

Problem C

For the marginal query,

$$f(Y = 1) = (0.5)(0.8) + (0.5)(0.2) = 0.5, \qquad f(Y = 0) = 0.5,$$
$$\mathsf{P}(Z = 1) = (0.5)(0.9) + (0.5)(0.1) = \boxed{0.50}.$$

For the posterior query, eliminate Y after entering $Z = 1$:

$$h(1) = (0.8)(0.9) + (0.2)(0.1) = 0.74, \qquad h(0) = (0.2)(0.9) + (0.8)(0.1) = 0.26.$$

The unnormalized weights are $u(1) = (0.5)(0.74) = 0.37$ and $u(0) = (0.5)(0.26) = 0.13$. Hence,

$$\mathsf{P}(X = 1 \mid Z = 1) = \frac{0.37}{0.37 + 0.13} = \boxed{0.74}.$$

Study reminders

- In Naive Bayes, independence is conditional on the class. In the binary word-presence model, absent words must also be considered.
- A zero feature likelihood makes the corresponding product zero. For categorical data, Laplace smoothing replaces a count estimate with $(n_{i,v,c} + 1)/(N_c + K_i)$, where K_i is the number of possible values of feature i and N_c is the number of class c examples.
- In Variable Elimination, sum over hidden variables; keep the query variables. Restrict factors to any observed evidence.
- A correct calculation is exact for its model; the model's assumptions and probability estimates still determine how well it describes reality.